

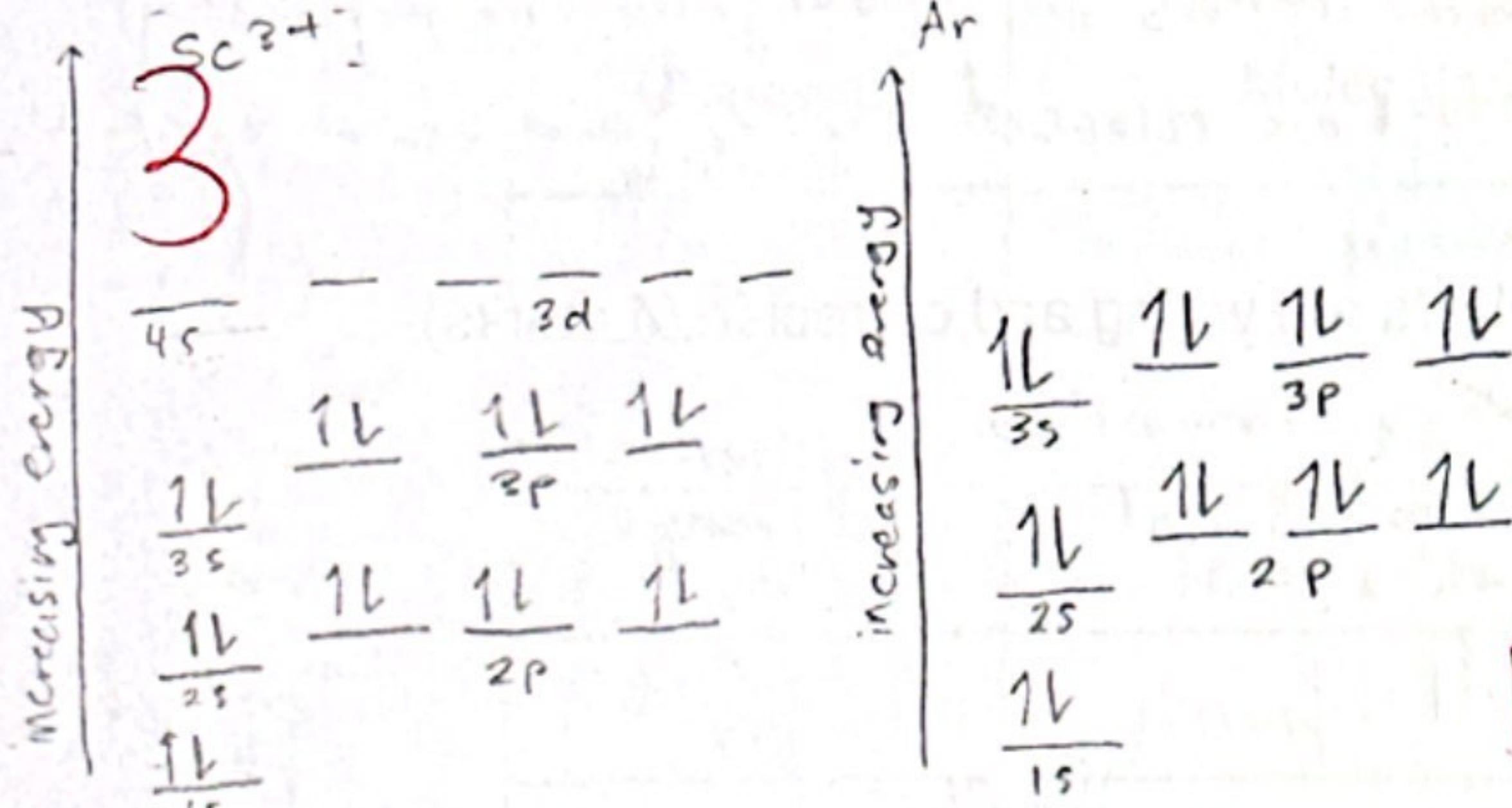
SCH 4U1 CHAPTER 3 QUIZ - ATOMIC THEORIES NAME: _____

PART B: SHORT ANSWER QUESTIONS

Answer the following questions in the space provided.

Inquiry:

1. Draw an energy level diagram for a scandium 3+ ION and argon atom. How do these diagrams compare? (3 marks).



The diagrams are ^{almost} identical as scandium with 3 less electrons, has the same amount as argon. ($21 - 3 = 18$)

And since cations have their valence electrons taken from the highest levels first, the diagrams stop at the 3p orbital.

Beautiful! They are "iso-electric"

2. Use shorthand electron configurations to explain why tin can have a charge of +2 and +4. Make sure to write the shorthand electron configuration for tin atom. (3 marks)

Sn: $[Kr] 5s^2 4d^{10} 5p^2$ ✓

When atoms lose their electrons to become an ion, they always forfeit the electrons in their highest energy level orbit.

Meaning that by losing 2 electrons, the 5p orbital is empty and by losing 4 electrons, the 5s orbital also becomes empty.

Since empty orbits are more stable than half-filled ones, with only a full 4d orbital left, tin ions become stable.

Sn^{2+} : $[Kr] 5s^2 4d^{10}$ ✓

Sn^{4+} : $[Kr] 4d^{10}$ ✓

3. Write the full electron configuration for selenium AND list the 4 quantum numbers for the last electron you drew in the configuration. Show all work. (3 marks)

Se: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$ ✓

last electron is $4p^4$

energy level = 4 → $n = 4$

$l = n - 1$ $4 - 1 = 3$ (max l value)

0 1 2 3 $l = 1$

s p d f

$m_l: -l \rightarrow l$ $(-1, 0, 1)$ $\frac{1}{-1}$ $\frac{1}{4p}$ $\frac{1}{+1}$ $m_l = -1$

$m_s: 4p^4$ is a downspin electron

∴ $m_s = -\frac{1}{2}$

4, 1, -1, $-\frac{1}{2}$ ✓

4. Use only shorthand electron configurations diagrams to prove that a molybdenum atom has an anomalous electron configuration. (HINT: Show a comparison with molybdenum's predicted and actual electron configuration) (3 marks)

Expected:

$[Kr] \frac{1}{5s} \frac{1}{4d} \frac{1}{4d} \frac{1}{4d} \frac{1}{4d}$

Actual:

$[Kr] \frac{1}{5s} \frac{1}{4d} \frac{1}{4d} \frac{1}{4d} \frac{1}{4d} \frac{1}{4d}$

↑ promotion

Since the stability of the d orbital is so great, it takes one electron from the 5s orbital to fill every orbital with a lone electron rather than having an incomplete 4d orbital

* label arrow as promotion

the intensity of the laser beam would be low since it would not be able to build up enough (a large #) of photons

Application:

★ include word "photon"

1. What would most likely happen to the intensity of a laser beam if the mirrors in the device were misaligned? Explain your answer. (2 marks)

The intensity will lower since the mirrors help create more charged photons by clashing them into each other electrons. If mirrors are misaligned, the number of photons generated will be less as the photons are released earlier and can't create more of them.

2. Explain why the following sets of quantum numbers are wrong and correct it. (4 marks)

a) 2, 1, 3, + 1/2

→ m_l ranges from $-l \rightarrow +l$ meaning that 3 isn't in that range

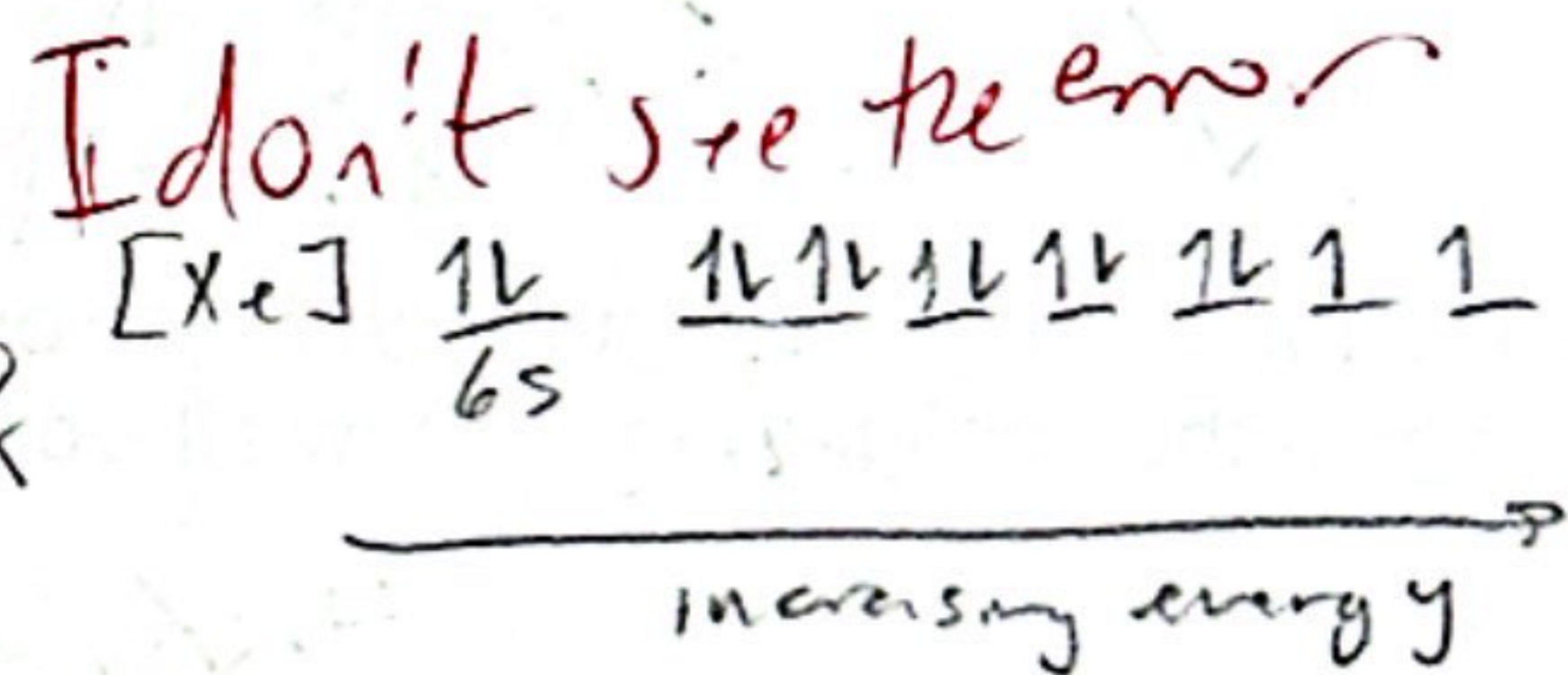
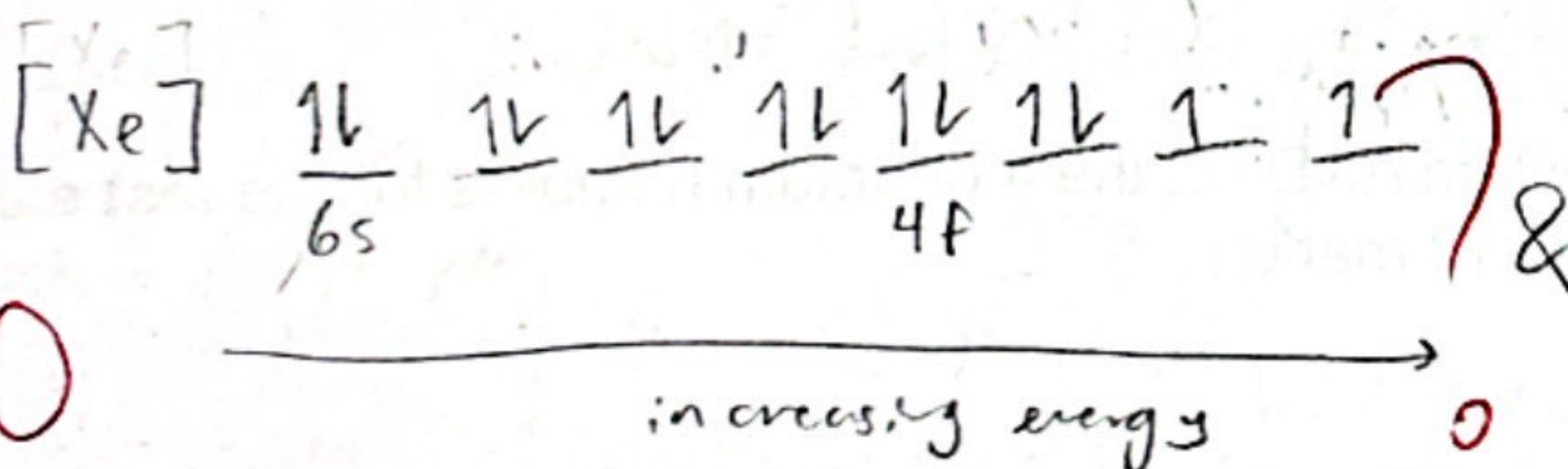
m_l needs to be $-1, 0, +1$

b) 1, 1, 1, - 1/2

→ the highest l value is equal to $n-1$. In this case, $n=1$ meaning either l must be changed to 0 and m_l along side it ($-l \rightarrow +l$), or we can increase the n number to 2 or higher.

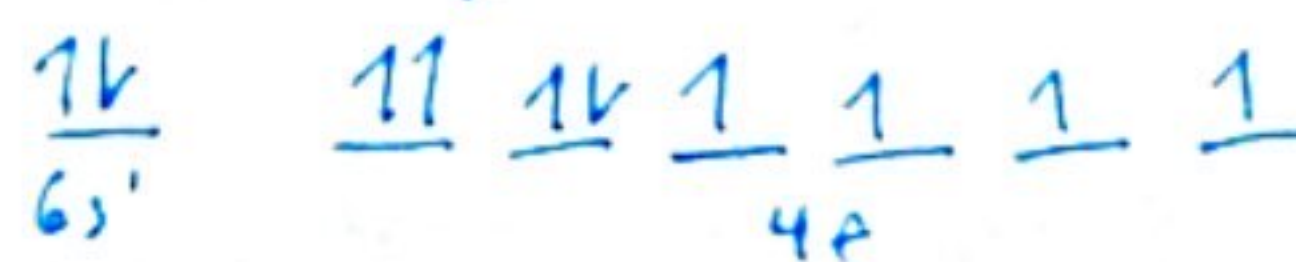
3. Draw a shorthand electron configuration diagram using the 4f orbitals that (2 marks)

(a) shows a violation of Pauli Exclusion's principle

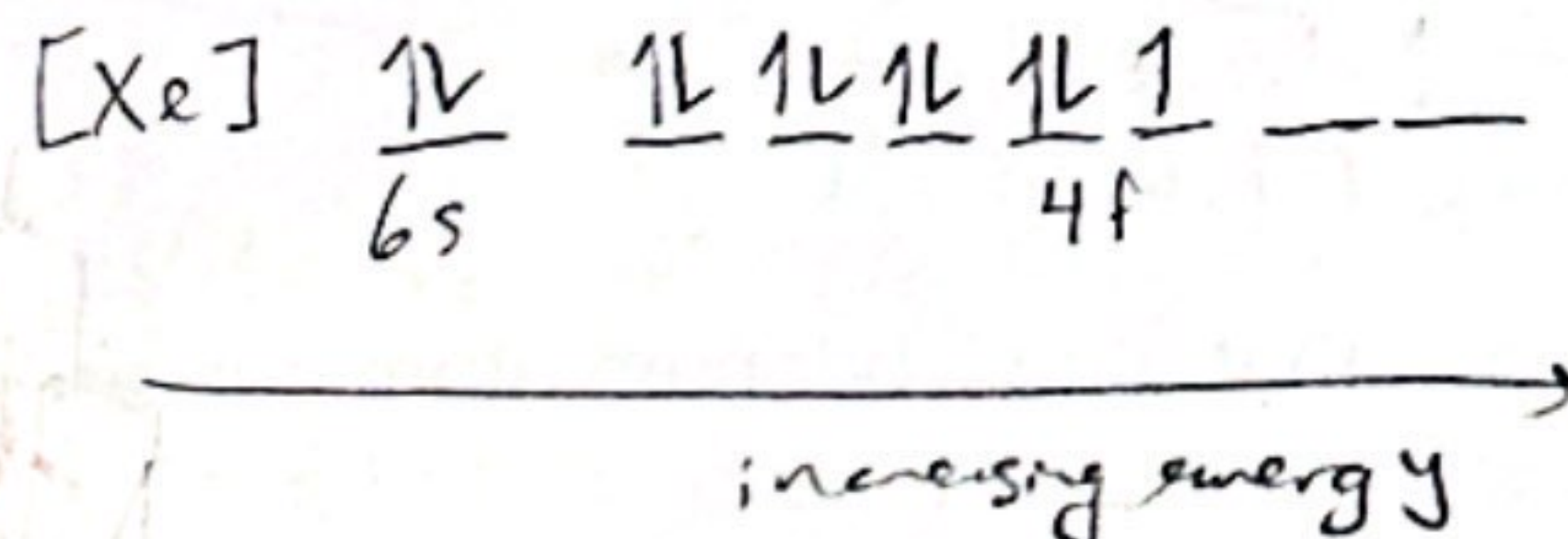


Both on SAME ATOM FOR DIFF ELECTRONS

↓ can't have the SAME SPIN



(b) shows a violation of the Hund's rule



Terbium (Tb)

4. Identify the element whose highest energy electron would have the following four quantum numbers: (2 marks)

6, 0, 0, +1/2

(Cs) cesium

5, 3, -3, -1/2 (Cm) curium